Qi Wu

Optical Metrology · Surface Science · Instrumentation

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Profile. Researcher and hands-on engineer with 2,000+ cleanroom hours and a strong track record of surface modification and optical measurement systems R&D. Cross-disciplinary background across optics, MEMS, and electrical engineering, with deep familiarity with metrology and characterization tools (ATR-FTIR, ellipsometry, SEM, AFM) and a structured, root-cause approach to measurement reliability.

EDUCATION

Ph.D. in Chemistry — *KU Leuven & imec, Belgium | 2022–Feb.2027 (expected)*

M.Sc. in Electrical Engineering — *University of Twente, the Netherlands | 2019–2021*

B.Eng. in Electrical Engineering — *Northwestern Polytechnical University, China | 2015–2019*

TECHNICAL SKILLS

Characterization & Metrology: FTIR spectroscopy; ellipsometry; wettability analysis; SEM; AFM.

Instrumentation & Prototyping: CAD design and build of custom metrology systems; re-engineering of commercial instruments; developing embedded microcontrollers; 3D-printing prototyping; measurement automation.

Optical Simulation: FDTD simulation of evanescent-field and near-field optics; spectral correction, processing, and analysis.

Programming & Data: Python and MATLAB for data processing, analysis, and hardware-software integration.

Experiment & Process: design of experiments (DOE); self-assembled monolayers (SAMs) LPCVD; wet-chemical surface processing; 2,000+ hrs cleanroom operation experience.

Languages: English (professional), Dutch (beginner), Mandarin Chinese (native).

RESEARCH EXPERIENCE

PhD Researcher — *imec, Leuven, Belgium | 2022–Present*

Superhydrophobic Surface Fabrication by Silane SAM Deposition

- **Surface engineering:** engineered superhydrophobic silicon surfaces via optimized silane self-assembled monolayer (SAM) deposition on patterned substrates, reaching intrinsic and apparent water contact angles above 110° and 140°.
- **Mechanistic analysis:** resolved the deposition mechanism by GATR-FTIR spectroscopy, profiling the kinetic balance between surface -OH loss during HF pre-cleaning and re-oxidation during CVD.
- **Process control:** established a robust pre-cleaning and deposition SOP with over 90% run-to-run reproducibility; recognized with the Best Student Paper Award at UCPSS 2025.

Quantitative Optical Metrology of Wetting Defects in HAR Nanopatterns

- **Instrument development:** built a customized ATR module for a commercial FTIR platform with an adjustable optical path, tuning the number of IR reflections to raise signal-to-noise ratio while avoiding absorption saturation.
- **Structure-to-spectrum mapping:** established a quantitative relation between nanopattern geometry (height 50-800 nm, surface fraction 0.1-0.8, at fixed 90 nm pitch) and ATR-FTIR absorbance, yielding a forward model from which the air-trapped fraction is inverted from a measured spectrum.
- **Physical verification of the model:** the near-field distribution at the structured interface under total internal reflection was computed by FDTD, confirming that the measured absorbance change originates from the filled trench geometry.
- **Validity domain:** defined the geometric window where the proportionality between air-trapping rate and absorbance loss remains valid (height lower than 500 nm, surface fraction less than 0.7); the two bounds were traced to the finite penetration depth and to breakdown of the total internal reflection condition.

Synchronized Cross-View Imaging Platform for Anisotropic Wetting Characterization

- **Platform design:** designed and built a cross-view optical platform for simultaneous, dual-directional imaging of sessile droplets on nano-trenches.

- **Measurement accuracy:** eliminated the timing mismatch of sequential imaging, enabling synchronized capture of anisotropic droplet elongation with improved morphological accuracy.
- **New defect indicator:** identified the structural-parameter regime where conventional wetting models break down and established real-time anisotropic elongation as a new indicator of nanoscale wetting defects.

Graduate Intern *ARCNL, Amsterdam, the Netherlands | 2021*

- **Mechanical simulation:** ran stress-strain simulations (COMSOL) of elastomer microstructure arrays for low-wear wafer carriers.
- **Industrial collaboration:** worked with cleanroom specialists to improve micro-component durability in a research-to-manufacturing setting.

PUBLICATIONS & PRESENTATIONS

Publications

- Q. Wu, et al., *"Modulating Silane Monolayer Grafting on Nanopatterned Silicon: Insights from Infrared-Spectroscopy-Based Air-Trapping Monitoring,"* **Solid State Phenomena** (UCPSS 2025, 17th International Symposium on Ultra Clean Processing of Semiconductor Surfaces), https://doi.org/10.4028/p-D9VeJm.

Presentations

- Q. Wu, *"Impact of HF Pretreatment on Silane Self-Assembled Monolayers on Silicon Surfaces,"* **oral presentation**, 18th Mini-Symposium on Liquids, Okayama, Japan, May 2025.
- Q. Wu, *"Optical Method to Characterize the Wetting States on Nano-patterned Surfaces,"* **poster presentation**, SPP 2171 Summer School, Munster, Germany, July 2024.

INTERESTS

Tennis, bouldering, hiking, DIY electronics.